

Gate Valve



Valve Standard

Conforms to AWWA C515/BS5163

Working Pressure

300 psi (21 bar)

Max Working Temperature

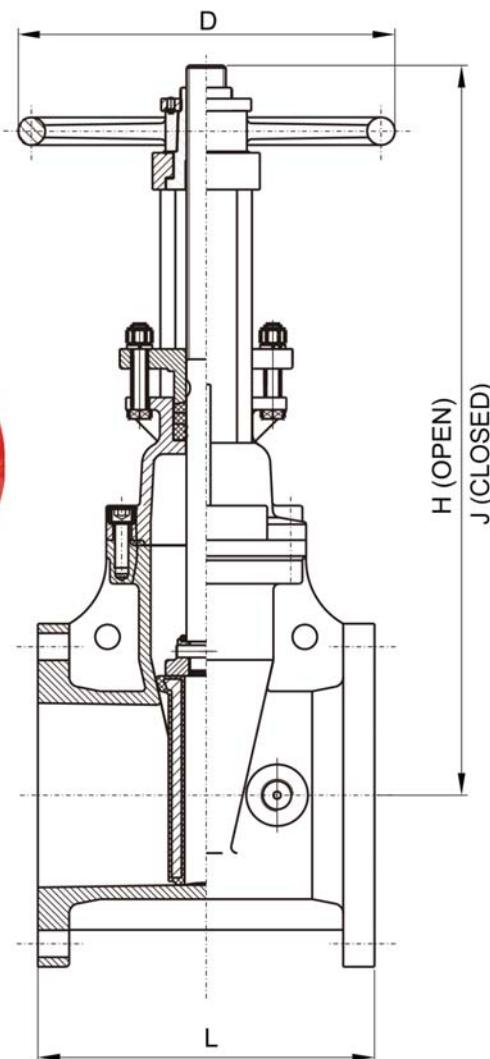
230°F (110°C)

Flanged Ends

ANSI B16.1 Class 125

Materials List

Part Name	Material	Specification
Body	Ductile Iron	A536 65-45-12
Wedge	Ductile Iron EPDM Coated	
Wedge Nut	Stainless Steel	AISI304
Stem	Stainless Steel	AISI304/302
Bonnet	Ductile Iron	A536 65-45-12
Gasket	EPDM	Commercial
Packing	Graphite	Non-asbestos
Stem Nut	Bronze	ASTM B62
Handwheel	Ductile Iron	A536 65-45-12



Dimensions: Inch (mm)

Size	2 ½ (65)	3 (80)	4 (100)	5 (125)	6 (150)	8 (200)	10 (250)	12 (300)
L	7.5 (191)	8.0 (203)	9.0 (229)	10.0 (254)	10.5 (267)	11.5 (292)	13.0 (330)	14.0 (356)
H	16.46 (418)	19.21 (488)	21.18 (538)	25.75 (654)	29.13 (740)	36.81 (935)	44.33 (1126)	51.81 (1316)
J	13.8 (350.5)	15.96 (405.5)	17.17 (436)	20.79 (528)	23.27 (591)	28.74 (730)	34.29 (871)	39.8 (1011)
D	7.2 (183)	9.96 (253)	9.96 (253)	12.05 (306)	12.05 (306)	13.98 (355)	17.52 (445)	17.52 (445)

Weight: lbs (kg)

Size	2 (50)	2 ½ (65)	3 (80)	4 (100)	5 (125)	6 (150)	8 (200)	10 (250)	12 (300)
W	41.89 (19)	46.3 (21)	61.73 (28)	81.57 (37)	110.23 (50)	127.87 (58)	216.05 (98)	295.42 (134)	418.88 (190)

Working Pressure & Temperature

Max Working Pressure:	300PSI / 21 Bar
Working Temperature:	-10°C to 110°C (14°F to 230°F)
	Fusion bonded epoxy coated interior and exterior RAL3002
Flanged Ends:	ANSI B16.1 Class 125/150/250 EN1092-2 PN10/PN16/PN25 BS10
Valve Standard:	Conforms to AWWA C508, Clear Waterway



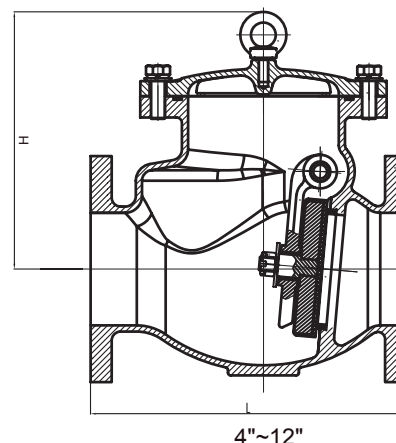
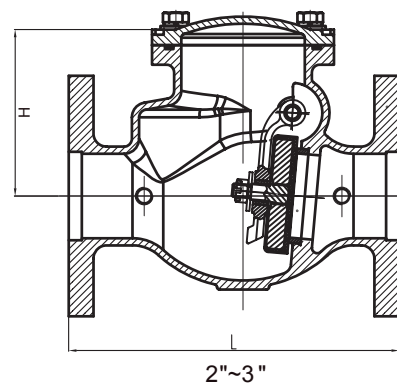
Material List

Component	Material
Body	Ductile Iron ASTM A536 65-45-12
Cover	Ductile Iron ASTM A536 65-45-12
Clapper	Ductile Iron ASTM A536 65-45-12 +EPDM
Clapper Arm	Ductile Iron ASTM A536 65-45-12
Hinge Pin	Stainless Steel AISI 304
Seat	Bronze ASTM B62
Seat	Stainless Steel AISI 304

Dimensions Inch (mm) and Weight lbs (kg)

DN	L	H	Weight
2 (50)	8.00 (203)	4.60 (117)	23.81 (10.8)
2 ½ (65)	10.00 (254)	5.04 (128)	34.72 (15.75)
3 (80)	11.00 (279)	5.21 (132.5)	43.65 (19.8)
4 (100)	13.00 (330)	8.76 (222.5)	67.46 (30.6)
5 (125)	14.00 (356)	11.42 (290)	115.08 (52.2)
6 (150)	16.00 (406)	11.69 (297)	133.53 (60.57)
8 (200)	19.50 (495)	14.37 (365)	238.1 (108)
10 (250)	22.00 (559)	16.34 (417.5)	345.24 (156.6)
12 (300)	26.00 (660)	17.83 (453)	488.1 (221.4)

Design and materials subject to change without notice.



Ordering Information

Description	Size (mm)	Part Number
RASCO, (PN16) FxF, 300PSI FCV, L501F	50 (2")	7U99002363
RASCO, (PN16) FxF, 300PSI FCV, L501F	65 (2 ½")	7U99002364
RASCO, (PN16) FxF, 300PSI FCV, L501F	80 (3")	7U99002365
RASCO, (PN16) FxF, 300PSI FCV, L501F	100 (4")	7U99002366
RASCO, (PN16) FxF, 300PSI FCV, L501F	125 (5")	7U99002367
RASCO, (PN16) FxF, 300PSI FCV, L501F	150 (6")	7U99002368
RASCO, (PN16) FxF, 300PSI FCV, L501F	200 (8")	7U99002369
RASCO, (PN16) FxF, 300PSI FCV, L501F	250 (10")	7U99002370
RASCO, (PN16) FxF, 300PSI FCV, L501F	300 (12")	7U99002371

Description	Size (mm)	Part Number
RASCO, (ANSI) FxF, 300PSI FCV, L501F	50 (2")	7U99002372
RASCO, (ANSI) FxF, 300PSI FCV, L501F	65 (2 ½")	7U99002373
RASCO, (ANSI) FxF, 300PSI FCV, L501F	80 (3")	7U99002374
RASCO, (ANSI) FxF, 300PSI FCV, L501F	100 (4")	7U99002375
RASCO, (ANSI) FxF, 300PSI FCV, L501F	150 (6")	7U99002376
RASCO, (ANSI) FxF, 300PSI FCV, L501F	200 (8")	7U99002377
RASCO, (ANSI) FxF, 300PSI FCV, L501F	250 (10")	7U99002378
RASCO, (ANSI) FxF, 300PSI FCV, L501F	300 (12")	7U99002379

EN 331/98

R 250D Giacomini ball valve

The R250D lever ball valve has been carefully designed and tested in order to offer a product of high quality, having safety features over and above most other ball valves making it extremely difficult to beat. One particular feature is the assembly of the operating stem. Most ball valves have the stem assembled from the outside (this kind of design is forbidden in many countries, especially if the valve is being used with dangerous fluids such as natural or town gas). For obvious safety reasons the stem of the Giacomini valve is assembled from

the inside of the valve thus protecting it from vandalism or sabotage. The stem is held secure by a special anti-friction washer. It cannot be withdrawn and judging by experiences throughout the world it is a safe and sound precaution recommended when used in domestic premises. The valve has been designed so that loss of pressure is minimal however a graph is available for each size of valve. Developed following full and comprehensive laboratory tests our valves comply with the requirements of international standard institutes.

Features

From 1/4" to 3/4":

- working with water, oil, gas 600 psi (42 kg/cm²)
- working with saturated steam up to 150 psi (10,5 kg/cm²) corresponds a temperature of 185°C.

From 1" and higher dimensions:

- 500 psi (35 kg/cm²) for water, oil, gas
- 150 psi (10,5 kg/cm²) for saturated steam at 185°C.

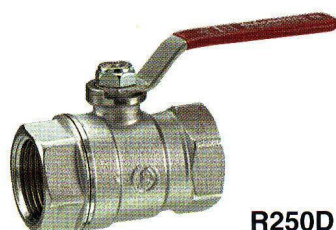
Technical data

The ball valve is in forged brass nickel and chromium plated.
The ball is nickel and chromium

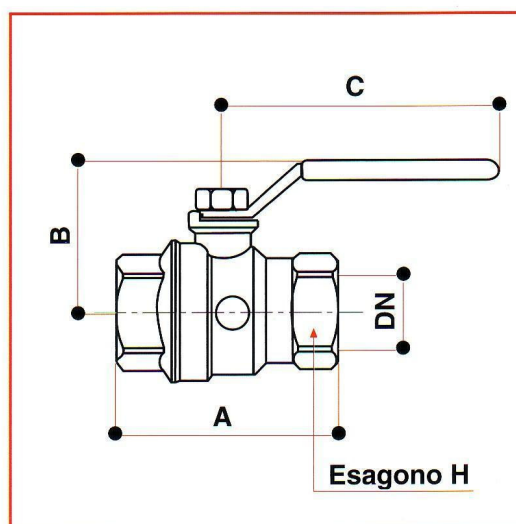
plated after diamond finishing.
All moving metal parts have teflon seals.

Seal on the spindle: 2 PTFE washers
1 viton O-Ring

Materials



R250D

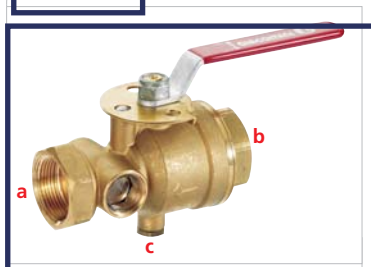


Dimensioni

	DN	A	B	C	H	KV
1/4"	8	39	26	42	17	6,6
3/8"	10	45	32	78	21	6,7
1/2"	15	54	34	78	25	10,2
3/4"	20	62	45	96	32	18,5
1"	25	75	50	96	39	36,3
1 1/4"	32	84	54	96	48	73,5
1 1/2"	40	93	68	138	54	105
2"	50	107	74	138	67	158
2 1/2"	65	143	101	172	83	240
3"	80	160	109	172	98	269
4"	100	203	131	165	123	461

Test And Drain Valve

A61



► FEMALE x FEMALE

Standard equipment:
Female NPT inlet and outlet, forged brass body, chrome plated ball with teflon seats.

Rated pressure: 300 psi

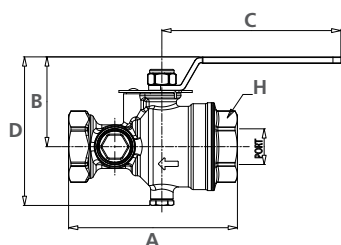


FM APPROVAL ONLY FOR 1" and 1-1/4"

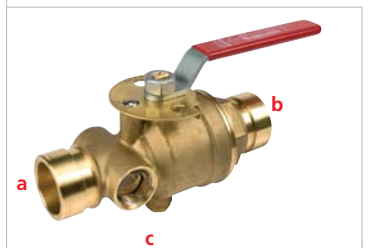
a/b	1"	1"	1"	1"	1-1/4"	1-1/4"	1-1/4"	1-1/4"	1-1/4"	1-1/4"
Orifice	3/8"	7/16"	1/2"	17/32"	3/8"	7/16"	1/2"	17/32"	5/8"	3/4"
K factor	2.8	4.2	5.6	8.0	2.8	4.2	5.6	8.0	11.2	14.0

a/b	2"							
Orifice	7/16"	1/2"	17/32"	5/8"	3/4"	15/16"	1-5/64"	1-9/64"
K factor	4.2	5.6	8.0	11.2	14.0	16.8	22.4	25.2

DIMENSION	C (plug)	PORT	A	B	C	D	H
1" & 1-1/4"	1/4"	1-1/16"	5-1/32"	2-43/64"	5-11/32"	4-27/64"	1-57/64"
2"	1/4"	1-12/16"	6-6/32"	4"	6-26/32"	6-21/64"	2-40/64"



A61



► GROOVE X GROOVE

Standard equipment:
GROOVE inlet and outlet, forged brass body, chrome plated ball with teflon seats.

Rated pressure: 300 psi

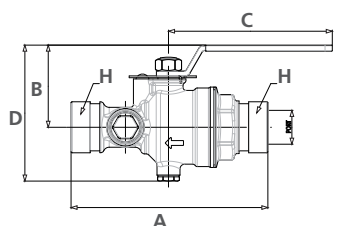


FM APPROVAL ONLY FOR 1" and 1-1/4"

a/b	1-1/4"					
Orifice	3/8"	7/16"	1/2"	17/32"	5/8"	3/4"
K factor	2.8	4.2	5.6	8.0	11.0	14.0

a/b	2"							
Orifice	7/16"	1/2"	17/32"	5/8"	3/4"	15/16"	1-5/64"	1-9/64"
K factor	4.2	5.6	8.0	11.2	14.0	16.8	22.4	25.2

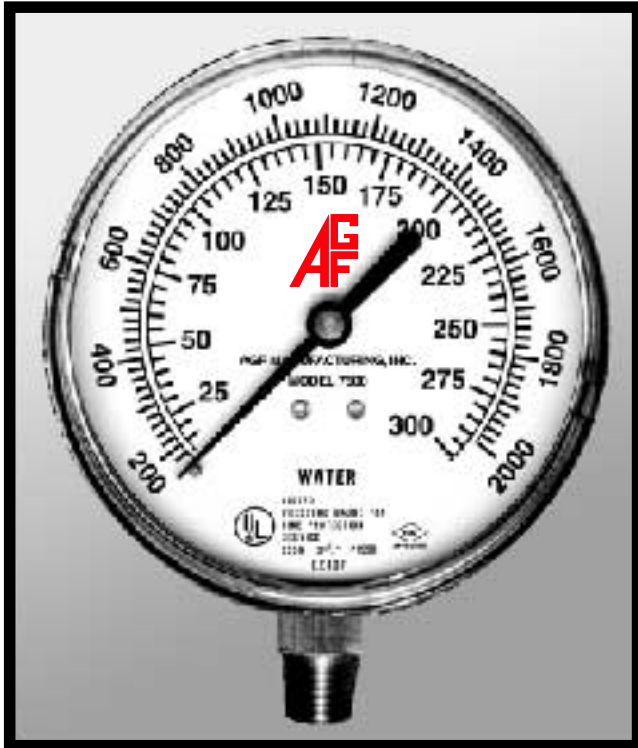
DIMENSION	C (plug)	PORT	A	B	C	D	H
1-1/4"	1/4"	1-1/16"	6-13/32"	2-43/64"	5-11/32"	4-27/64"	1-1/4" GROOVE"
2"	1/4"	1-12/16"	7-17/32"	4"	6-26/32"	6-21/64"	2" GROOVE"





MODEL 7500

PRESSURE GAUGE **FOR FIRE SPRINKLER SYSTEMS**



The AGF Manufacturing, Inc. **Model 7500** Pressure Gauge is a Factory Mutual Approved and Underwriters Laboratory Listed 3 1/2" gauge designed for use on "wet" sprinkler systems. The AGF **Model 7500** has been designed to meet NFPA System Attachments requirements as specified in NFPA 13 Sec. 4-7.3. The **Model 7500** fulfills the installation specifications for a pressure gauge as stipulated in NFPA 13 Sec. 3-1.1, 3-7.7, 4-6.1.2.2, NFPA 13D Sec. 3-2, and NFPA 13R Sec. 2-4.3.

SPECIFICATIONS

RANGE

0-300 psi (inside)
0-2000 Kpa (outside)

ACCURACY

±3-2-3%

BURDON TUBE

Phosphor Bronze

WINDOW

Polycarbonate one-piece ring and crystal

DIAL

Galvalume

POINTER

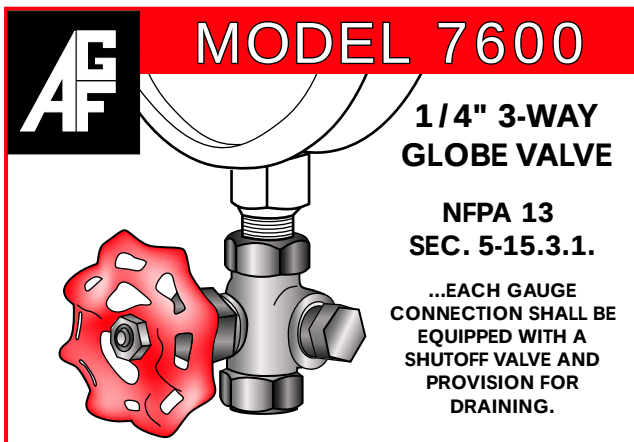
Aluminum, painted black

CASE

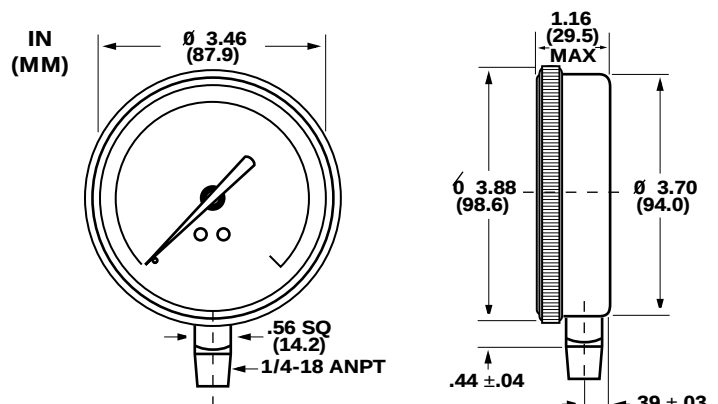
PGE Xenoy® DX2735

CONNECTION

Brass, 1/4" ANPT LM



DIMENSIONS



AGF Manufacturing, Inc. provides many different products to address a variety of job conditions. Please contact our office for advice and products modified to solve specific job conditions. Thank you for using our products.

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AGF MANUFACTURING INC.

100 Quaker Lane
Malvern, PA 19355 USA
Telephone: (610) 240-4900
Fax: (610) 240-4906
www.testandrain.com

JOB NAME: _____
ARCHITECT: _____
ENGINEER: _____
CONTRACTOR: _____



Specifications subject to change without notice.

Ordering Information			
Nominal Pipe Size		Model	Part Number
2"	DN50	VSR-2	1144402
2 1/2"	DN65	VSR-2 1/2	1144425
3"	DN80	VSR-3	1144403
3 1/2"	-	VSR-3 1/2	1144435
4"	DN100	VSR-4	1144404
5"	-	VSR-5	1144405
6"	DN150	VSR-6	1144406
8"	DN200	VSR-8	1144408

Optional: Cover Tamper Switch Kit, stock no. 0090148

Replaceable Components: Retard/Switch Assembly, stock no. 1029030

UL, CUL and CSFM Listed, FM Approved, LPCB Approved, For CE Marked (EN12259-5) / VdS Approved model use VSR-EU

Service Pressure: 450 PSI (31 BAR) - UL

Flow Sensitivity Range for Signal:

4-10 GPM (15-38 LPM) - UL

Maximum Surge: 18 FPS (5.5 m/s)

Contact Ratings: Two sets of SPDT (Form C)
10.0 Amps at 125/250VAC
2.0 Amps at 30VDC Resistive
10 mAmps min. at 24VDC

Conduit Entrances: Two knockouts provided for 1/2" conduit.
Individual switch compartments suitable for dissimilar voltages.

Environmental Specifications:

- NEMA 4/IP54 Rated Enclosure suitable for indoor or outdoor use with factory installed gasket and die-cast housing when used with appropriate conduit fitting.
- Temperature Range: 40°F - 120°F, (4.5°C - 49°C) - UL
- Non-corrosive sleeve factory installed in saddle.

Service Use:

Automatic Sprinkler	NFPA-13
One or two family dwelling	NFPA-13D
Residential occupancy up to four stories	NFPA-13R
National Fire Alarm Code	NFPA-72

WARNING

- Installation must be performed by qualified personnel and in accordance with all national and local codes and ordinances.
- Shock hazard. Disconnect power source before servicing. Serious injury or death could result.
- Risk of explosion. Not for use in hazardous locations. Serious injury or death could result.

CAUTION

Waterflow switches that are monitoring wet pipe sprinkler systems shall not be used as the sole initiating device to discharge AFFF, deluge, or chemical suppression systems. Waterflow switches used for this application may result in unintended discharges caused by surges, trapped air, or short retard times.

General Information

The Model VSR is a vane type waterflow switch for use on wet sprinkler systems. It is UL Listed and FM Approved for use on steel pipe; schedules 10 through 40, sizes 2" thru 8" (50 mm thru 200 mm). LPC approved sizes are 2" thru 8" (50 mm thru 200 mm). See Ordering Information chart.

The VSR may also be used as a sectional waterflow detector on large systems. The VSR contains two single pole, double throw, snap action switches and an adjustable, instantly recycling pneumatic retard. The switches are actuated when a flow of 10 GPM (38 LPM) or more occurs downstream of the device. The flow condition must exist for a period of time necessary to overcome the selected retard period.

Enclosure

The VSR switches and retard device are enclosed in a general purpose, die-cast housing. The cover is held in place with two tamper resistant screws which require a special key for removal. A field installable cover tamper switch is available as an option which may be used to indicate unauthorized removal of the cover. See bulletin number 5401103 for installation instructions of this switch.

Installation (see Fig. 1)

These devices may be mounted on horizontal or vertical pipe. On horizontal pipe they shall be installed on the top side of the pipe where they will be accessible. The device should not be installed within 6" (15 cm) of a fitting which changes the direction of the waterflow or within 24" (60 cm) of a valve or drain.

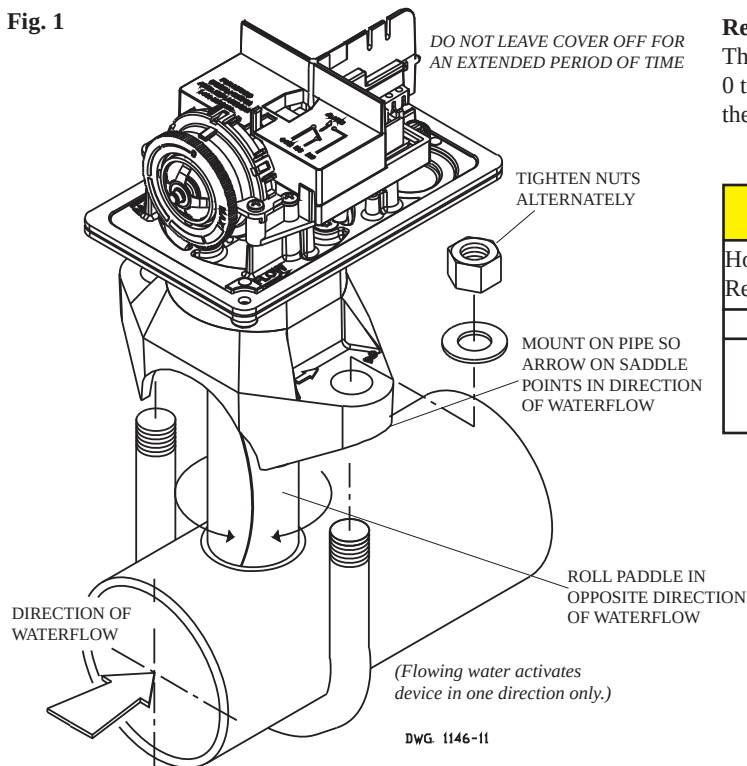
NOTE: Do not leave cover off for an extended period of time.

Drain the system and drill a hole in the pipe using a hole saw in a slow speed drill (see Fig. 1). Clean the inside pipe of all growth or other material for a distance equal to the pipe diameter on either side of the hole. Roll the vane so that it may be inserted into the hole; do not bend or crease it. Insert the vane so that the arrow on the saddle points in the direction of the waterflow. Take care not to damage the non-corrosive bushing in the saddle. The bushing should fit inside the hole in the pipe. Install the saddle strap and tighten nuts alternately to required torque (see the chart in Fig. 1). The vane must not rub the inside of the pipe or bind in any way.

CAUTION

Do not trim the paddle. Failure to follow these instructions may prevent the device from operating and will void the warranty.

Fig. 1

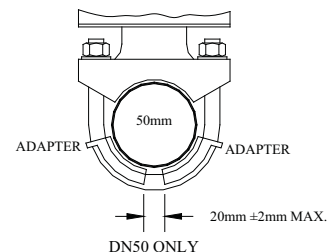
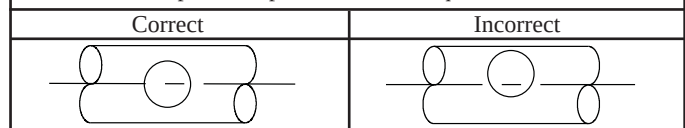


Retard Adjustment

The delay can be adjusted by rotating the retard adjustment knob from 0 to the max setting (60-90 seconds). The time delay should be set at the minimum required to prevent false alarms

CAUTION

Hole must be drilled perpendicular to the pipe and vertically centered. Refer to the Compatible Pipe/Installation Requirements chart for size.



DN50 ONLY

USE (2) 5180162 ADAPTERS AS SHOWN ABOVE
DWG# 1146-1F

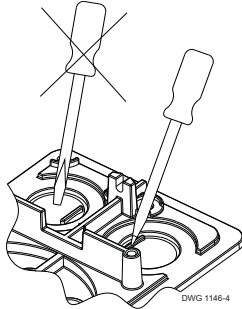
Compatible Pipe/ Installation Requirements

Model	Nominal Pipe Size		Nominal Pipe O.D.		Pipe Wall Thickness								Hole Size		U-Bolt Nuts Torque	
					Schedule 10 (UL)		Schedule 40 (UL)		BS-1387 (LPC)		DN (VDS)					
	inch	mm	inch	mm	inch	mm	inch	mm	inch	mm	inch	mm	inch	mm	ft-lb	n-m
VSR-2	2	DN50	2.375	60.3	0.109	2.77	0.154	3.91	0.142	3.6	0.091	2.3	1.25 + .125/-0.062	33.0 ± 2.0	20	27
VSR-2 1/2	2.5	-	2.875	73.0	0.120	3.05	0.203	5.16	-	-	-	-				
VSR-2 1/2	-	DN65	3.000	76.1	-	-	-	-	0.142	3.6	0.102	2.6				
VSR-3	3	DN80	3.500	88.9	0.120	3.05	0.216	5.49	0.157	4.0	0.114	2.9	2.00 ± .125	50.8 ± 2.0		
VSR-3 1/2	3.5	-	4.000	101.6	0.120	3.05	0.226	5.74	-	-	-	-				
VSR-4	4	DN100	4.500	114.3	0.120	3.05	0.237	6.02	0.177	4.5	0.126	3.2				
VSR-5	5	-	5.563	141.3	0.134	3.40	0.258	6.55	-	-	-	-				
VSR-6	6	DN150	6.625	168.3	0.134	3.40	0.280	7.11	0.197	5.0	0.157	4.0				
VSR-8	8	DN200	8.625	219.1	0.148	3.76	0.322	8.18	0.248	6.3	0.177	4.5				

NOTE: For copper or plastic pipe use Model VSR-CF.

Fig. 2

To remove knockouts: Place screwdriver at inside edge of knockouts, not in the center.



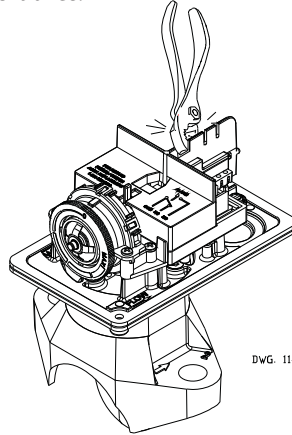
DWG. 1146-4

NOTICE

Do not drill into the base as this creates metal shavings which can create electrical hazards and damage the device. Drilling voids the warranty.

Fig. 3

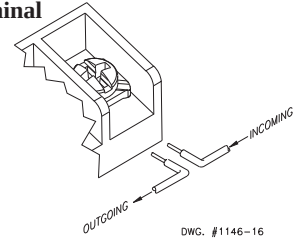
Break out thin section of cover when wiring both switches from one conduit entrance.



DWG. 1146-13

Fig. 4

Switch Terminal Connections Clamping Plate Terminal



DWG. #1146-16

WARNING

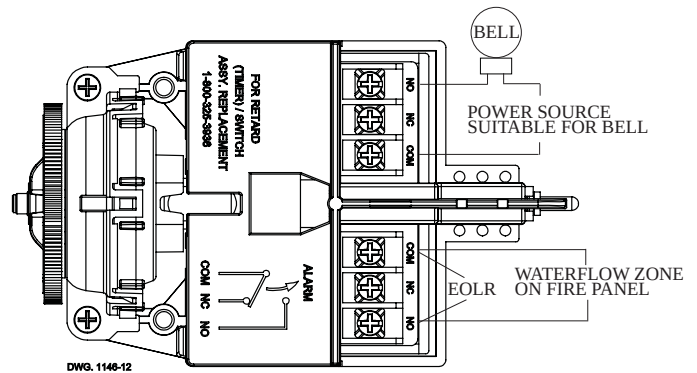
An uninsulated section of a single conductor should not be looped around the terminal and serve as two separate connections. The wire must be severed, thereby providing supervision of the connection in the event that the wire become dislodged from under the terminal. Failure to sever the wire may render the device inoperable risking severe property damage and loss of life.

Do not strip wire beyond 3/8" of length or expose an uninsulated conductor beyond the edge of the terminal block. When using stranded wire, capture all strands under the clamping plate.

Fig. 5 Typical Electrical Connections

Notes:

1. The Model VSR has two switches, one can be used to operate a central station, proprietary or remote signaling unit, while the other contact is used to operate a local audible or visual annunciator.
2. A condition of LPC Approval of this product is that the electrical entry must be sealed to exclude moisture.
3. For supervised circuits, see "Switch Terminal Connections" drawing and warning note (Fig. 4).



DWG. 1146-12

Testing

The frequency of inspection and testing for the Model VSR and its associated protective monitoring system shall be in accordance with applicable NFPA Codes and Standards and/or the authority having jurisdiction (manufacturer recommends quarterly or more frequently).

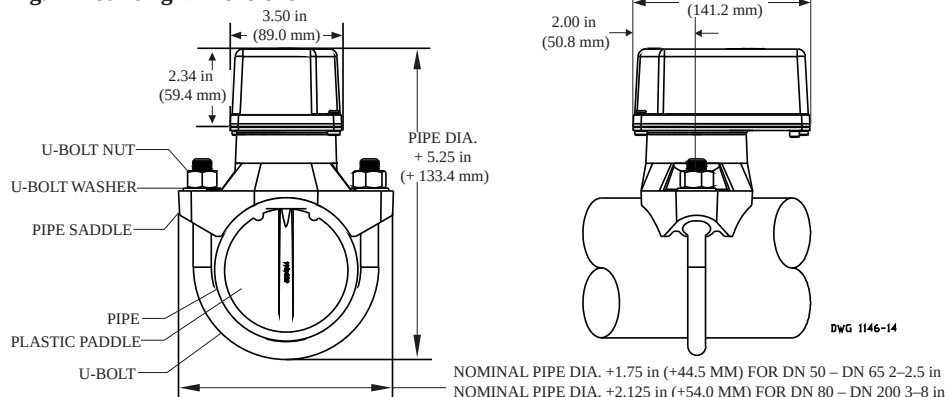
If provided, the inspector's test valve shall always be used for test purposes. If there are no provisions for testing the operation of the flow detection device on the system, application of the VSR is not recommended or advisable.

A minimum flow of 10 GPM (38 LPM) is required to activate this device.

NOTICE

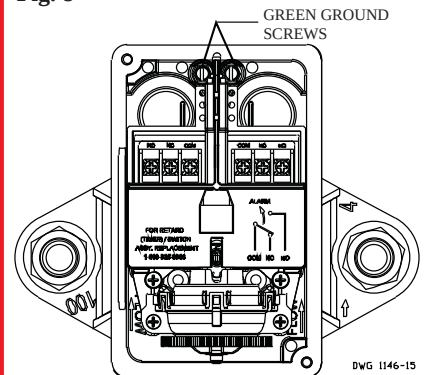
Advise the person responsible for testing of the fire protection system that this system must be tested in accordance with the testing instructions.

Fig. 7 Mounting Dimensions



DWG. 1146-14

Fig. 8



DWG. 1146-15

Maintenance

Inspect detectors monthly. If leaks are found, replace the detector. The VSR waterflow switch should provide years of trouble-free service. The retard and switch assembly are easily field replaceable. In the unlikely event that either component does not perform properly, please order replacement retard switch assembly stock #1029030 (see Fig. 6). There is no maintenance required, only periodic testing and inspection.

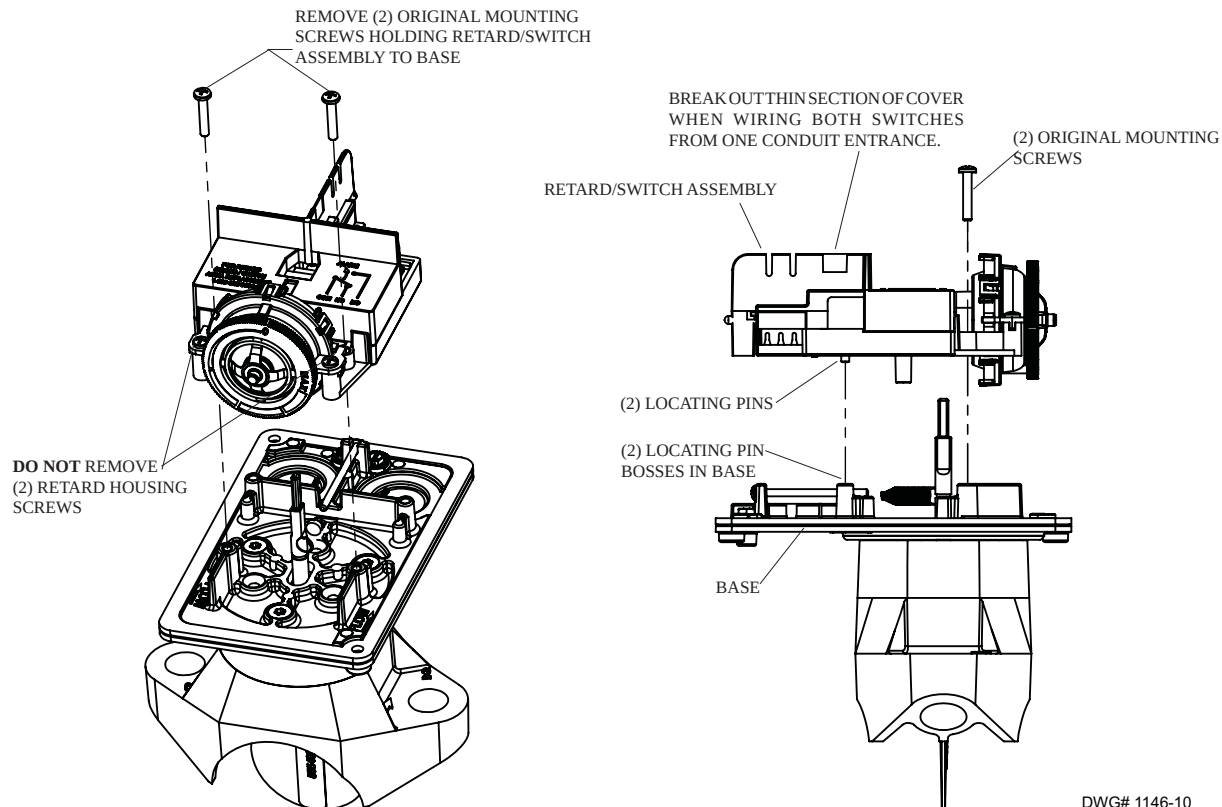
Retard/Switch Assembly Replacement (See Fig. 6)

NOTICE

The Retard/Switch Assembly is field-replaceable without draining the system or removing the waterflow switch from the pipe.

1. Make sure the fire alarm zone or circuit connected to the waterflow switch is bypassed or otherwise taken out of service.
2. Disconnect the power source for local bell (if applicable).
3. Identify and remove all wires from the waterflow switch.
4. Remove the (2) mounting screws holding retard/switch assembly to the base. **Do not** remove the (2) retard housing screws.
5. Remove the retard assembly by lifting it straight up over the tripstem.
6. Install the new retard assembly. Make sure the locating pins on the retard/switch assembly fit into the locating pin bosses on the base.
7. Re-install the (2) original mounting screws.
8. Reconnect all wires. Perform a flow test and place the system back in service.

Fig. 6



Removal of Waterflow Switch

- To prevent accidental water damage, all control valves should be shut tight and the system completely drained before waterflow detectors are removed or replaced.
- Turn off electrical power to the detector, then disconnect wiring.
- Loosen nuts and remove U-bolts.
- Gently lift the saddle far enough to get your fingers under it. With your fingers, roll the vane so it will fit through the hole while continuing to lift the waterflow detector saddle.
- Lift detector clear of pipe.